# Answer key

**Practice Test 2 (State level)** · 50 questions · 33 with checked app keys

Each answer lists short steps, the vault note (Question Types folder) and the app keys. Every key line was run in the app and its screen value matched the computed answer (check_keys.py). Accept numeric answers within about 2% with the right unit.

### Multiple-choice answers

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| D | B | B | B | C | D | B | A | D | A | B | B | A | D |

| 15 | 16 |
|---|---|
| A | C |

## Section I: Multiple choice

**1. D. A university student**

1. Kirchhoff stated the laws in 1845 as a student (about age 21).
2. They generalize Ohm's law (1827) to circuits with many loops.

Note: A1 History/Kirchhoff
App: no app: fact or concept

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**2. B. A torsion balance**

1. Coulomb used a torsion balance (1785).
2. The twist of a thin fiber measured very small forces.

Note: A1 History/Coulomb
App: no app: fact or concept

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**3. B. The electricity came from two different metals in contact**

1. Volta said two different metals with a moist conductor make the electricity.
2. This idea led to his voltaic pile (1800).

Note: A1 History/Volta
App: no app: fact or concept

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**4. B. Making and detecting electromagnetic (radio) waves**

1. Hertz made radio waves with a spark gap and detected them with a loop.
2. This confirmed Maxwell's theory.

Note: A1 History/Hertz
App: no app: fact or concept

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**5. C. Spread apart, because they become negative**

1. The rod pushes electrons away from the knob, down to the leaves.
2. Both leaves become negative and repel.
3. The knob becomes positive; the total charge stays zero.

Note: A2 Static Electricity and Coulomb/Charging Methods and Electroscopes
App: no app: fact or concept

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**6. D. On the outer surface**

1. Like charges push apart as far as they can.
2. In a conductor they end up on the outer surface.
3. The field inside the metal is zero.

Note: A2 Static Electricity and Coulomb/Electric Field and Potential
App: no app: fact or concept

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**7. B. AC makes muscles clamp, so the person cannot let go**

1. AC near 60 Hz makes muscles contract again and again (tetany).
2. The hand can clamp on the wire, so the shock lasts longer.

Note: A3 DC and AC/Shock and Safety
App: no app: fact or concept

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**8. A. 1.5 V and 4000 mAh**

1. Parallel: the voltage stays 1.5 V.
2. The capacities add: 4000 mAh.

Note: A3 DC and AC/Batteries and Cells
App: no app: fact or concept

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**9. D. Much too low**

1. The meter is in parallel with the resistor.
2. 10 kohm || 1 Mohm is about 9.9 kohm.
3. The measured part now has much less resistance, so it takes a much smaller voltage.

Note: A5 DC Circuits/Meters and Measurement
App: no app: fact or concept

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**10. A. It can be reset after it trips**

1. A fuse melts and must be replaced.
2. A breaker opens and can be reset (switched back on).

Note: A6 Switches Relays Fuses Breakers GFCI/Circuit Breakers
App: no app: fact or concept

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**11. B. A person on a rubber mat touches the hot and the neutral wire at the same time**

1. A GFCI trips on a difference between hot and neutral currents.
2. Current that goes in on hot and back on neutral stays balanced.
3. So the GFCI cannot see it.

Note: A6 Switches Relays Fuses Breakers GFCI/GFCI
App: no app: fact or concept

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**12. B. XOR (or XNOR)**

1. Flipping either switch changes the lamp.
2. That is XOR or XNOR, depending on how 'up' is labeled.

Note: A6 Switches Relays Fuses Breakers GFCI/3-Way and 4-Way Switching
App: no app: fact or concept

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**13. A. n-type; electrons**

1. Phosphorus has 5 valence electrons (a donor).
2. The extra electron is free: n-type.
3. The majority carriers are electrons.

Note: A7 Diodes and LEDs/Semiconductors and Doping
App: no app: fact or concept

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**14. D. L1 only**

1. Conventional current leaves the + side (top).
2. D1 points down, the way the current flows: forward bias.
3. D2 points up, against the current: reverse bias, it blocks.
4. Only L1 lights.

Note: A7 Diodes and LEDs/Diode Circuit Analysis
App: no app: fact or concept

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**15. A. Sum = XOR, Carry = AND**

1. 1 + 1 = 10: Sum 0, Carry 1.
2. Sum is 1 when exactly one input is 1: XOR.
3. Carry is 1 only when both are 1: AND.

Note: A8 Digital Logic/Adders
App: no app: fact or concept

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**16. C. 3**

1. A + B = (A'B')' by De Morgan.
2. Two NANDs make A' and B' (inputs tied together).
3. A third NAND gives (A'B')' = A + B.

Note: A8 Digital Logic/Universal Gates (NAND and NOR)
App: `t-(-(aa)-(bb))=a+b` → `EQUAL (all 4 rows match)` *(checked)*

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## Section II: Short answer and calculations

**17. 4.0 N, attractive**

1. $F = kq_1q_2/r^2$ with r = 0.15 m.
2. F = 8.99e9 × 2.5e-6 × 4.0e-6 / $0.15^2$ = 4.0 N.
3. Opposite signs attract. The app shows a negative force = attraction.

Note: A2 Static Electricity and Coulomb/Coulombs Law
App: `q a2.5u b-4u r.15` → `f Force (+ repel) = -3.994 N` *(checked)*

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**18. 400 mN, toward –x (left)**

1. From +3.0 μC: attraction to the left, k(3.0e-6)(2.0e-6)/$0.30^2$ = 0.599 N.
2. From +1.0 μC: attraction to the right, k(1.0e-6)(2.0e-6)/$0.30^2$ = 0.200 N.
3. Net = 0.599 − 0.200 = 400 mN, to the left.

Note: A2 Static Electricity and Coulomb/Coulombs Law
App: `Q 3u 0, -2u .3, 1u .6` → `F on q2 (-2 uC at 0.3): -399.4 mN left` *(checked)*

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**19. E = 860 N/C, V = 220 V**

1. E = kq/$r^2$ = 8.99e9 × 6.0e-9 / 0.0625.
2. E = 860 N/C, pointing away.
3. V = kq/r = 220 V.

Note: A2 Static Electricity and Coulomb/Electric Field and Potential
App: `f q6n r.25` → `e Field E = 862.8 N/C` *(checked)*
App: `f q6n r.25` → `v Potential V = 215.7 V` *(checked)*

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**20. 3 Ω**

1. Lost volts = 9.0 – 8.1 = 0.9 V (1 significant figure).
2. r = 0.9 V / 0.30 A = 3 Ω.

Note: A5 DC Circuits/Internal Resistance and Power Transfer
App: `i e9 v8.1 i.3` → `r Internal r = 3 ohm` *(checked)*

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**21. 230 V**

1. $V_{rms} = V_{peak} / \sqrt{2}$.
2. 325 / 1.414 = 230 V.

Note: A3 DC and AC/AC Characteristics
App: `w p325` → `r RMS = 229.8 V` *(checked)*

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**22. 2.14 m**

1. A = πd$^2$/4 = 1.963e-7 m$^2$.
2. L = RA/ρ = 12.0 × 1.963e-7 / 1.10e-6 = 2.14 m.

Note: A4 Current Voltage Resistance Power Energy/Resistance and Resistivity
App: `r r12 p1.1E-6 d.5m` → `l Length = 2.142 m` *(checked)*

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**23. 5.4 MJ (= 1.5 kWh)**

1. t = 45 × 60 = 2700 s.
2. E = Pt = 2000 × 2700 = 5.4 MJ.
3. In kWh: 2.0 kW × 0.75 h = 1.5 kWh.

Note: A4 Current Voltage Resistance Power Energy/Power Energy and Cost
App: `J p2000 t45min` → `j Energy (joules) = 5.4 MJ` *(checked)*

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**24. 20.4 W; the 40 W bulb is brighter**

1. R40 = 120$^2$/40 = 360 ohm; R100 = 144 ohm.
2. I = 120 / 504 = 238 mA.
3. P40 = $I^2$R = 20.4 W.
4. Series: same I, so the larger R gets more power.

Note: A5 DC Circuits/Dividers and Bulb Brightness
App: `B40 120, 100 120 =120` → `bulb 1: 85.71 V, P = 20.41 W` *(checked)*

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**25. 3.90 kΩ, ±1%**

1. orange 3, white 9, black 0: 390.
2. brown multiplier = 10: 3900 ohm.
3. brown tolerance = ±1%.

Note: A5 DC Circuits/Resistor Color Codes
App: `c owknn` → `= 3900 ohm` *(checked)*
App: `c owknn` → `3.9 kohm +/-1%` *(checked)*

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**26. 450 Ω**

1. Balanced: R1/R2 = R3/$R_x$.
2. $R_x$ = R3 × R2 / R1 = 150 × 360 / 120 = 450 Ω.

Note: A5 DC Circuits/Hard Resistor Networks
App: `W a120 b360 c150` → `x Rx (bottom right) = 450 ohm` *(checked)*

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**27. 4.76 V**

1. Meter || 100 kohm = 100k × 1M / 1.1M = 90.9 kohm.
2. V = 10.0 × 90.9 / (100 + 90.9) = 4.76 V.

Note: A5 DC Circuits/Meters and Measurement
App: `d v10 a100k b100k/1M` → `o Vout (across R2) = 4.762 V` *(checked)*

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**28. 20 mA**

1. $V_R$ = 5.0 − 0.7 = 4.3 V.
2. I = 4.3 / 220 = 20 mA.

Note: A7 Diodes and LEDs/Diode Circuit Analysis
App: `l s5 f.7 r220` → `i LED current = 19.55 mA` *(checked)*

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**29. 250 Ω**

1. Both LEDs drop 2 × 2.0 = 4.0 V.
2. $V_R$ = 9.0 − 4.0 = 5.0 V.
3. R = 5.0 / 0.020 = 250 Ω.

Note: A7 Diodes and LEDs/LEDs
App: `l s9 f2 c2 i20m` → `r Resistor = 250 ohm` *(checked)*

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**30. F = AB + C**

1. The four terms with C cover every AB: they make C.
2. ABC' + ABC = AB.
3. F = AB + C.

Note: A8 Digital Logic/Boolean Algebra and Simplification
App: `t-a-bc+-abc+a-bc+ab-c+abc` → `F = AB + C` *(checked)*

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### 31. F = A′B′C + A′BC′ + AB′C′ + ABC; a 3-input XOR

1. Write one product for each row with F = 1: 001, 010, 100, 111.
2. F = 1 when an odd number of inputs are 1.
3. That is A XOR B XOR C.

Note: A8 Digital Logic/Truth Tables and Boolean Expressions
App: `t01101001` → `F = A'B'C + A'BC' + AB'C' + ABC` *(checked)*
App: `t01101001` → `This is a 3-input XOR gate` *(checked)*

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### 32. 1001000 = 72

1. 101101 = 45; 11011 = 27.
2. 45 + 27 = 72.
3. 72 = 64 + 8 = 1001000.

Note: A8 Digital Logic/Binary Numbers and Addition
App: `x101101+11011` → `sum bin 1001000` *(checked)*

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### 33. Y = S′$I_0$ + S$I_1$

1. S = 0 passes I0: term S'I0.
2. S = 1 passes I1: term SI1.
3. OR the two terms.

Note: A8 Digital Logic/Multiplexers and Decoders
App: by hand (concept)

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### 34. 32 mW

1. P = VI.
2. P = 9.0 × 0.0036 = 32 mW.

Note: A9 Hands-On and Meters/Task - Power Supplied
App: `o v9 i3.6m` → `p Power = 32.4 mW` *(checked)*

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### 35. (1.0k + 470) || (330 + 220) = 400 Ω

1. Try two series pairs in parallel.
2. 1470 || 550 = 1470 × 550 / 2020.
3. = 400 Ω (0.06% high).

Note: A9 Hands-On and Meters/Task - Target Resistance
App: `h 220 330 470 1k=400` → `400.2 (+0.0619%) (1k+470)||(330+220)` *(checked)*

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### 36. 400 Ω (the red one needs 467 Ω)

1. $V_R$ = 9.0 − 3.0 = 6.0 V.
2. R = 6.0 / 0.015 = 400 Ω.
3. Red: (9.0 − 2.0)/0.015 = 467 ohm.

Note: A9 Hands-On and Meters/Task - Equal LEDs
App: `l s9 f3 i15m` → `r Resistor = 400 ohm` *(checked)*
App: `l s9 f2 i15m` → `r Resistor = 466.7 ohm` *(checked)*

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## Section III: Circuit problems

**Problem A: two batteries (Kirchhoff's laws) (Questions 37–42)**

E1 = 12.0 V, E2 = 6.0 V, R1 = 4.0 Ω, R2 = 2.0 Ω, R3 = 6.0 Ω. Both batteries have their + side at the top. Junction X is at the top of R3 and junction Y at the bottom. Ideal batteries and wires. Let $I_a$ be the clockwise current in the left loop and $I_b$ the clockwise current in the right loop.

### 37. $10I_a - 6I_b = 12$ and $-6I_a + 8I_b = -6$

1. Left loop: E1 = R1 Ia + R3 (Ia − Ib): 12 = 4Ia + 6(Ia − Ib).
2. Right loop, clockwise: going down through E2 from + to − is a drop of 6 V.
3. 0 = 2Ib + 6 + 6(Ib − Ia), so −6Ia + 8Ib = −6.

Note: A5 DC Circuits/Kirchhoff Analysis (States-Nationals)
App: by hand (write the equations); solve them with the j keys in the next question

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### 38. 1.36 A

1. Solve: det = 10 × 8 − 36 = 44.
2. $I_a$ = (12 × 8 − 36)/44 = 1.36 A.
3. R1 carries $I_a$ (left to right).

Note: A5 DC Circuits/Kirchhoff Analysis (States-Nationals)
App: `j a10 b-6 c12 d-6 e8 f-6` → `x x (I1 or V1) = 1.3636` *(checked)*
App: `N V1-0 12 1-2 4 2-0 6 V3-0 6 3-2 2` → `R1-2 4: 5.455 V 1.364 A 1>2 7.438 W` *(checked)*

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### 39. 273 mA; E2 is being charged (current enters its + side)

1. $I_b$ = (10 × (−6) + 72)/44 = 273 mA.
2. $I_b$ > 0 clockwise: it flows down into the + side of E2.
3. Current into the + side means E2 takes energy: it is charged.

Note: A5 DC Circuits/Kirchhoff Analysis (States-Nationals)
App: `j a10 b-6 c12 d-6 e8 f-6` → `y y (I2 or V2) = 0.27273` *(checked)*
App: `N V1-0 12 1-2 4 2-0 6 V3-0 6 3-2 2` → `V3-0 6: 272.7 mA into +, takes 1.636 W` *(checked)*
App: `N V1-0 12 1-2 4 2-0 6 V3-0 6 3-2 2` → `V3-0 6: 272.7 mA into +, takes 1.636 W` *(checked)*

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### 40. 1.09 A, down (X to Y)

1. I3 = $I_a - I_b$ = 1.364 − 0.273 = 1.09 A.
2. Positive, so it flows down from X to Y.
3. KCL check at X: 1.364 in = 1.091 + 0.273 out.

Note: A5 DC Circuits/Kirchhoff Analysis (States-Nationals)
App: `N V1-0 12 1-2 4 2-0 6 V3-0 6 3-2 2` → `R2-0 6: 6.545 V 1.091 A 2>0 7.14 W` *(checked)*

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### 41. 6.55 V

1. $V_{XY}$ = I3 × R3 = 1.091 × 6.0 = 6.55 V.
2. Check from the left: 12.0 − 1.364 × 4.0 = 6.55 V.

Note: A5 DC Circuits/Kirchhoff Analysis (States-Nationals)
App: `N V1-0 12 1-2 4 2-0 6 V3-0 6 3-2 2` → `V1 = 12 V V2 = 6.545 V` *(checked)*

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**42. 16.4 W**

1. $P = E1 \times I_a = 12.0 \times 1.364 = 16.4$ W.

Note: A5 DC Circuits/Kirchhoff Analysis (States-Nationals)
App: `N V1–0 12 1–2 4 2–0 6 V3–0 6 3–2 2` → `V1–0 12: 1.364 A out of +, gives 16.36 W` *(checked)*

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**Problem B: unbalanced bridge (Questions 43–45)**

A 15.0 V battery is connected between A (+) and B (–). R1 = 100 Ω (A-C), R2 = 220 Ω (A-D), R3 = 330 Ω (C-B), R4 = 150 Ω (D-B), R5 = 470 Ω (C-D). The bridge is not balanced (R1/R3 is not equal to R2/R4).

**43. 4.01 V; C is higher**

1. Use B as 0 V. Unknowns: $V_C$, $V_D$ (two KCL equations).
2. At C: $(15 - V_C)/100 = V_C/330 + (V_C - V_D)/470$.
3. At D: $(15 - V_D)/220 + (V_C - V_D)/470 = V_D/150$.
4. $V_C = 10.86$ V, $V_D = 6.843$ V.
5. $V_C - V_D = 4.01$ V.

Note: A5 DC Circuits/Kirchhoff Analysis (States-Nationals)
App: `N V1–0 15 1–2 100 1–3 220 2–0 330 3–0 150 2–3 470 V2–3` → `V2 – V3 = 4.014 V` *(checked)*

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**44. 78.5 mA**

1. $I(R1) = (15 - V_C)/100 = 41.4$ mA.
2. $I(R2) = (15 - V_D)/220 = 37.1$ mA.
3. Total = 78.5 mA.

Note: A5 DC Circuits/Kirchhoff Analysis (States-Nationals)
App: `N V1–0 15 1–2 100 1–3 220 2–0 330 3–0 150 2–3 470` → `V1–0 15: 78.52 mA out of +, gives 1.178 W` *(checked)*

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**45. 191 Ω**

1. $R_{eq} = 15.0\text{ V} / 78.5\text{ mA} = 191\ \Omega$.

Note: A5 DC Circuits/Hard Resistor Networks
App: `N 1–2 100 1–3 220 2–0 330 3–0 150 2–3 470 =1–0` → `R(1–0) = 191 ohm` *(checked)*

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**Problem C: three identical bulbs (Questions 46–48)**

Three identical bulbs A, B and C are connected to an ideal battery as shown. Treat each bulb as a fixed resistor R. Brightness follows power.

**46. ×0.5625 = 9/16 (A gets dimmer)**

1. Before: $R_{eq} = R + R/2 = 1.5R$; $I_A = V/1.5R$.
2. After: $R_{eq} = 2R$; $I_A = V/2R$.
3. P ratio $= (1.5/2)^2 = 9/16 = 0.5625$: dimmer.

Note: A5 DC Circuits/Open and Short Circuits
App: `N V1–0 1 1–2 1 2–0 1 o2–0 1` → `1–2 1: 444.4 mW –> 250 mW x0.5625 dimmer` *(checked)*

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**47. ×2.25 (B gets brighter)**

1. Before: B has half of $I_A$: V/3R, so P = $V^2$/9R.
2. After: B carries V/2R, so P = $V^2$/4R.
3. Ratio = 9/4 = 2.25: brighter.

Note: A5 DC Circuits/Open and Short Circuits
App: `N V1–0 1 1–2 1 2–0 1 o2–0 1` → `2–0 1: 111.1 mW –> 250 mW x2.25 brighter` *(checked)*

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**48. A goes out; B and C each get 9 times their old power (much brighter)**

1. The wire has zero resistance, so A gets no current.
2. B and C now have the full V across each.
3. P = $V^2$/R, which is 9 times $V^2$/9R.

Note: A5 DC Circuits/Open and Short Circuits
App: `N V1–0 1 s1–2 1 2–0 1 2–0 1` → `2–0 1: 111.1 mW –> 1 W x9 brighter` *(checked)*

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**Problem D: logic circuit (Questions 49–50)**

Gate G1 is a NAND gate with inputs A and B. Gate G2 is an OR gate with inputs B and C. An AND gate combines G1 and G2 to make F.

**49. F = (AB)′(B + C) = A′B + B′C**

1. F = (AB)'(B + C) = (A' + B')(B + C) by De Morgan.
2. Expand: A'B + A'C + B'B + B'C.
3. B'B = 0. A'C is covered by A'B + B'C (consensus).
4. F = A'B + B'C.

Note: A8 Digital Logic/Boolean Algebra and Simplification
App: `t–(ab)(b+c)` → `F = A'B + B'C` *(checked)*

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**50. 001, 010, 011, 101**

1. A'B: rows 010, 011.
2. B'C: rows 001, 101.
3. Together: 001, 010, 011, 101.

Note: A8 Digital Logic/Truth Tables and Boolean Expressions
App: `t–(ab)(b+c)` → `m(1,2,3,5)` *(checked)*

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